

VELA

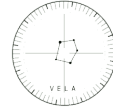
PRIVATE COMMAND INFRASTRUCTURE

OpenClaw Architecture Manual

System architecture, installation, agent design, and operations reference.

Your judgment. Our infrastructure.

Mac Mini M4 · OpenClaw 2026 · Version 2.0 · March 2026



01 Hardware

The Mac Mini M4 is the only supported hardware. Its unified memory architecture allows local AI inference to run alongside OpenClaw and all channel integrations without a discrete GPU.

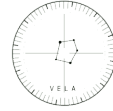
	32 GB / 1 TB (Recommended)	16 GB / 512 GB (Budget)
Price	~\$1,099	~\$599–799
All local models concurrent	Yes	7b only during active sessions
Resource contention risk	None	Low with correct config
Future-proof lifespan	4–5 years	2–3 years
VELA recommendation	Default recommend	Budget option — set expectations

Set Energy Saver to Never in System Settings. The Mac Mini must not sleep — session interruption during an active agent task corrupts the context file.

02 Software Stack

Component	Version	Purpose
macOS Sequoia	15.x	Host OS — Apple Silicon native
OpenClaw	2026.3.x	AI agent orchestration platform
Ollama	Latest via Homebrew	Local model inference server
Node.js	v22+	OpenClaw runtime dependency
Python	3.14	Automation scripts
gog-wrapper	Latest	Google Workspace CLI — Gmail, Calendar, Contacts

Provider	Models	Cost	When Used
Anthropic (Cloud)	claude-sonnet-4-6	\$3.00/MTok input	Hannah CoS — writing, strategy, judgment
Ollama (Local)	qwen2.5:7b, llama3.1:8b	\$0	PM, Researcher, email classification



03 Agent Architecture

Agent ID	Model	Cost per Task	Role
cos	claude-sonnet-4-6	~\$0.004–\$0.40	Chief of Staff — primary interface, orchestrator
analyst	claude-sonnet-4-6	~\$0.08–\$0.25	Financial modeling, scenario analysis
marketing	claude-sonnet-4-6	~\$0.08–\$0.25	Messaging, positioning, copy, brand voice
legal	claude-sonnet-4-6	~\$0.08–\$0.25	Contract review, risk flags, compliance
pm	ollama/qwen2.5:7b	\$0	Project tracking, milestones, commitments
researcher	ollama/qwen2.5:7b	\$0	Data reasoning over pre-fetched content

CRITICAL: Local agents (qwen2.5:7b) fail silently on tool execution. A failed local agent task triggers Hannah's retry loop which can cost \$5–10. Never assign tool-execution tasks to PM or Researcher.

Model Split Rationale — Validated by Real Data

12 days of real deployment data: model choice = 7.4% of total bill. Context management = 92.6%. Analyst, Marketing, and Legal on Sonnet costs ~\$0.08–0.25/task. Do NOT move all agents to local to save money — retry loop costs from failed tool execution exceed Sonnet costs by 10–50x.

04 Installation

The complete install runs from a single Terminal command. Time estimate: 30–90 minutes depending on model download speed.

```
curl -fsSL https://raw.githubusercontent.com/greg442/Vela_Scripts_2/main/install.sh | bash
```

The installer handles all 18 steps automatically:

- Homebrew and core dependencies
- Ollama + local models (15–30 min download)
- OpenClaw installation
- Directory structure creation
- API keys and environment configuration
- Telegram and WhatsApp configuration
- Google Workspace authentication
- Agent creation — 6 agents with correct model assignments
- Workspace assignment via Python JSON method (CLI cannot set this)
- Workspace template population with your client variables
- Automation scripts installation
- Cost optimization settings (cacheRetention: short)
- 5 scheduled cron jobs
- License validation active



05 Cost Model & Optimization

The Core Insight — Validated by Real Data

Output tokens were 7.4% of the total bill. Model choice is almost irrelevant to cost. Context management drives over 90% of API spend.

Token Type	Price per Million	Notes
Input (no cache)	\$3.00	Baseline
Cache write — 5-minute TTL	\$3.75	USE THIS — 1.25x base
Cache write — 1-hour TTL	\$6.00	NEVER USE — 2x base, was \$30.85 in 12 days
Cache read	\$0.30	Cheapest token in the system
Output	\$15.00	5x base — but only 7.4% of real-world bill

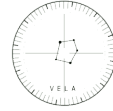
Scenario	Daily Cost	Notes
Light — briefs only, few tasks	\$1–\$3	Mostly cache reads + cron
Normal active day	\$3–\$6	Mix of input and cached context, 2–4 agent tasks
Heavy — multi-agent, long sessions	\$6–\$10	Multiple Sonnet spawns
Broken architecture (pre-fix)	\$11–\$27	Context bloat + wrong cacheRetention

06 Integrations

Telegram Configuration

Setting	Value	Notes
dmPolicy	pairing	Requires explicit pairing before DMs work
groupPolicy	allowlist	Only authorized group IDs can send commands
groupAllowFrom	[-GROUP_ID]	Numeric IDs only — not quoted strings
streaming	partial	Shows partial responses as they generate

Google Workspace (gog-wrapper)



```
# Gmail

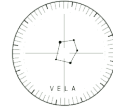
gog-wrapper gmail search "is:unread is:important" -a PRIMARY@gmail.com

gog-wrapper gmail send -a COS@gmail.com --to EMAIL --subject SUBJECT --body BODY


# Calendar

gog-wrapper calendar list -a PRIMARY@gmail.com --today

gog-wrapper calendar list -a PRIMARY@gmail.com --days 7
```



07 Automation Layer

Job	Schedule	Purpose
hannah-morning	Daily at {{BRIEF_MORNING_HOUR}}:00 {{CLIENT_TIMEZONE}}	Morning brief — email, calendar, priority surface
hannah-evening	Daily at {{BRIEF_EVENING_HOUR}}:00 {{CLIENT_TIMEZONE}}	Evening brief — day summary, tomorrow prep
email_triage.py	Every 15 min, weekdays 8am–6pm	Email triage — classify, flag, Telegram delivery
reset_sessions.sh	3am + 2pm daily	Delete session files over 100KB
cost_alert.py	6pm weekdays	Daily spend check — Telegram alert if elevated
backup_gdrive.sh	2am daily	Backup ~/.openclaw to Google Drive
license_check.py	9am daily	License validation — stand down if revoked

08 Operating Procedures & Troubleshooting

Problem	First Check	Fix
{{AGENT_NAME}} not responding	openclaw status	openclaw gateway restart
High API costs	Check cacheRetention + workspace map	Verify cacheRetention is short — never long
Specialist producing blank output	Run workspace map — check for (none)	Use Python JSON method to assign workspace
WhatsApp disconnected	openclaw logs --follow	Rescan QR codes via openclaw gateway restart
Session files bloated	ls ~/.openclaw/agents/cos/sessions/	Run reset_sessions.sh manually